Natural Products

Definition and Occurrence

- **A natural product** is a chemical compound or substance produced by living bodies found in nature that usually has a pharmacological or biological activity.
- Natural products chemistry, the chemistry of metabolic products of plants, animals and microorganisms, is involved in the investigation of biological phenomena in the human body.
- Natural product chemistry has become one of the most interesting and finest areas of modern chemistry as it is continuously driving the isolation of new bioactive compounds from living organisms for the remedy of fatal diseases.

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- Natural product is a chemical substance found in nature that has distinct biological and pharmacological effects. Natural product actually refers to as any naturally occurring substance, but is generally used to mean secondary metabolites which are small molecules that are not essential for the development of producing organism.
- Natural products may be extracted from tissues of terrestrial plants, animals, marine organisms or microorganisms. A crude extract from any one of these sources typically contains novel, structurally diverse chemical compounds, which the natural environment is a rich source of.

Importance of Some Natural Products

- Plants, which serve as source of energy and effective drugs, for example, the natural products such as vitamins, hormones, antibiotics, analgesics extracted from plants are very important for their medicinal uses.
- The importance of natural products and their applications in certain spheres of human life have attracted the Chemists and Druggists.

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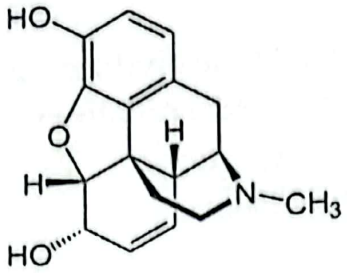
- As a result in the early decades of the civilization natural products chemistry achieved a revolutionary advancement leading to systematization to chemotherapy and this fruitful discovery has accelerated the research work for newer medicine.
- From time immemorial various traditional system of treatment of diseases like Kaviraze, Ayurvedic and Unani have developed in the world especially in the Indian Sub-continent based on the use of the medicinal plants. These medicinal plants were then only source to alleviate the sufferings of body and mind.

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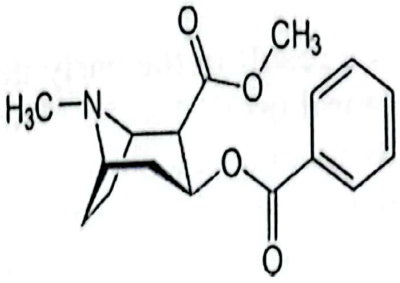
- With the development of modern chemistry and pharmacy at the beginning of the nineteenth century the original impetus in the study of natural products utilizing medicinal plants got a tremendous plants boost.
- Morphine, the hypnotic and anesthetic principle of opium; Quinine, the anti-malarial drug from cinchona bark and cocaine, used as local anesthetic from coca leaves were some examples in the list of early drugs. It was followed by the discovery of the Alkaloids used in the mental conditions including tension and anxiety as well as in the treatment of hypertension and more potent natural drug continued to be isolated.

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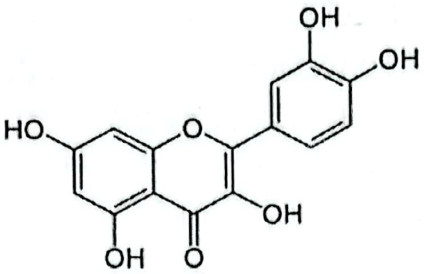
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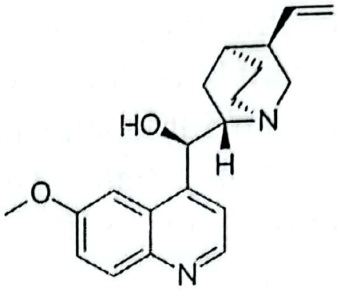
Morphine



Cocaine

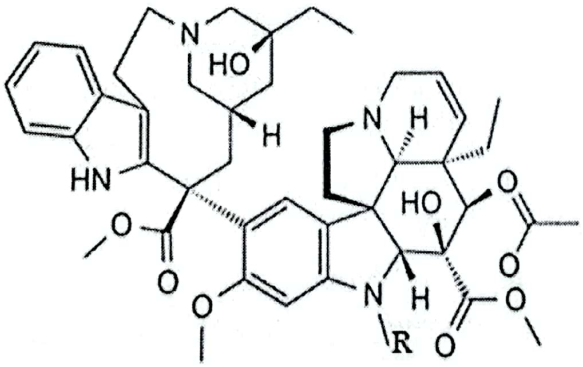


Quercetin

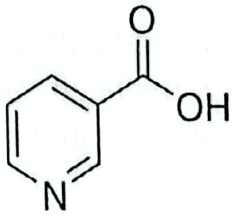


Quinine

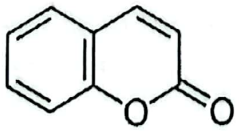
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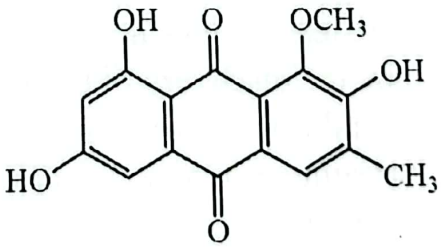
R = CHO (Vincristine)
R = CH_3 (Vinblastine)



Nicotinic acid



Coumarin



2-Hydroxyemodin-1-methyether

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- Two alkaloids: vinblastine and vincristine were isolated from the plant *catharanthus roseus* which are active against cancer.
- Coumarins have been isolated from many plants and have shown different pharmacological effects.
- 2-Hydroxyemodin-1-methylether isolated from *Ventilago leiocarpa* is used in the treatment of pain and rheumatism.
- Quercetin isolated from many plants is used as natural antioxidant.
- It is expected that natural product chemistry will provide more medicines for therapy of most fatal diseases like cancer and AIDS which remain to be conquered as yet.

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Terpenoids

Ref.: 1. Organic Chemistry, vol. 2 by I. L. Finar

2. Chemistry of Organic Natural Products by O. P. Agarwal

Definition

- Terpenoids are the compounds having the molecular formula $(C_5H_8)_n$ and their oxygenated derivatives like alcohols, aldehydes or ketones.
- Those have molecular formula $(C_5H_8)_n$ are called terpenoid hydrocarbons or terpenes. These are unsaturated hydrocarbons.

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Classification

Number of carbon atoms	Class	Mol. Formula of terpenes
10	Monoterpenoids	$C_{10}H_{16}$
15	Sesquiterpenoids	$C_{15}H_{24}$
20	Diterpenoids	$C_{20}H_{32}$
25	Sesterterpenoids	$C_{25}H_{40}$
30	Triterpenoids	$C_{30}H_{48}$
40	Tetraterpenoids	$C_{40}H_{64}$
>40	Polyterpenoids	$(C_5H_8)_n$

- Only very few sesterterpenoids are known. Carotenoids are tetraterpenoids (C_{40}).
- Each class of terpenoids has few sub-classes like acyclic, monocyclic, bicyclic, tricyclic, etc.
- Examples → Monoterpenoids: myrcene, Sesquiterpenoids: farnesol, Diterpenoids: Phytol, Sesterterpenoids: neomangicols A, Triterpenoids: squalene, Tetraterpenoids: β-carotene, Polyterpenoids: rubber.

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Occurrence

- The terpenoids form a group of compounds the majority of which occur in the plant kingdom; a few terpenoids have been obtained from other sources.
- The simpler mono- and sesquiterpenoids are the chief constituents of the essential oils; these are the volatile oils obtained from certain plants and trees.
- The di-, tri-, tetra- and polyterpenoids which are not steam volatile, are obtained from plant and tree gums and resins.

Importance

- Due to their pleasant smelling nature, the mono- and sesquiterpenoids are of considerable commercial importance, particularly in the perfumery industry.
- Various terpenoids show biological activity, e.g. insecticidal, anthelmintic or antiseptic action, and are thus also useful in pharmacy.
- Rubber is the most important polyterpenoid.

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Essential oils	Terpenoids
Jasmine	Linalool
Lemon	Limonene, Citral
Rose	Geraniol, Citronellol, Farnesol
Cardamon	Terpineol
Camphor	Camphor
Ginger	Zinziberene

Isolation and Purification

There are four methods of extraction of the terpenoids:

(i) Expression

(ii) Steam distillation

(iii) Extraction by means of volatile solvents

(iv) Adsorption in purified fats

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- Expression: Production of something by pressing or squeezing it out. Essential oils obtained by expression. The plant material is crushed and the juice is screened to remove the large particles. The screened juice is centrifuged in a high speed centrifugal machine when nearly half of the essential oil is extracted. Citrus, lemon and grass oils are extracted by this method.
- Method (ii) is the one most widely used; the plant is macerated and then steam distilled.

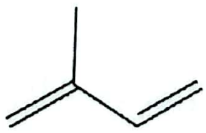
- If the compound decomposes under these conditions, it may be extracted with light petrol at 50 °C and the solvent then removed by distillation under reduced pressure.
- Alternatively, the method of adsorption in fat is used. The fat is warmed to about 50 °C, and then the flower petals are spread on the surface of the fat until the latter is saturated. The fat is now digested with ethanol, any fat that dissolves being removed by cooling to 20 °C.

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- The essential oils so obtained usually contain a number of terpenoids, and these are separated by fractional distillation. The terpenoid hydrocarbons distil first, and these are followed by the oxygenated derivatives.
- Distillation of the residue under reduced pressure gives the sesquiterpenoids, and these are separated by fractional distillation. Now-a-days, chromatography has been used both for isolation and separation of terpenoids.

Isoprene Rule

- The thermal decomposition of almost all terpenoids gives isoprene as one of the products, and this led to the suggestion that the skeleton structures of all naturally occurring terpenoids can be built up of isoprene units; this is known as the isoprene rule, and was first pointed out by Wallach (1887).

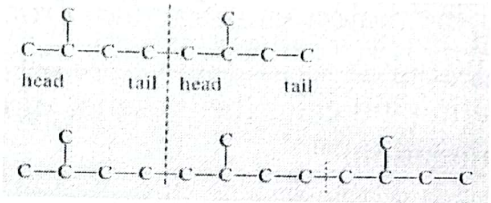


Isoprene
(2-Methyl-1,3-butadiene)

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Special Isoprene Rule

- Ingold (1925) pointed out that the isoprene units in natural terpenoids were joined 'head to tail' (the head being the branched end of isoprene). This divisibility into isoprene units, and their head to tail union, may conveniently be referred to as the special isoprene rule.
- Several exceptions occur, e.g., the carotenoids are joined tail to tail at their centre.



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General Methods of Determining Structure of Terpenoids

The following brief account gives an indication of the various methods which have been particularly useful in elucidating the structures of the terpenoids.

- A pure specimen is obtained, and the molecular formula is ascertained by the usual methods, and also by means of mass spectrometry.
- If the terpenoid is optically active, its specific rotation is measured.
- If oxygen is present in the molecule, its functional nature is ascertained, whether it is present as hydroxyl, aldehyde, ketone, etc.

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- The presence of olefinic bonds is ascertained by means of bromine, and the number of double bonds is determined by analysis of the bromide, or by quantitative hydrogenation. These facts lead to the molecular formula of the parent hydrocarbon, from which the number of rings present in the structure may be deduced.
- Dehydrogenation of the terpenoids with S, Se, Pt or Pd and an examination of the products thereby obtained. Measurement of the refractive index leads to a value for the molecular refraction. From this may be deduced the nature of the carbon skeleton. Also, optical exaltation indicates the presence of double bonds in conjugation.

- Degradative oxidation: The usual reagents used for this purpose are ozone, neutral or alkaline permanganate, chromic acid and sodium hypobromite. Other reagents are osmium tetroxide, nitric acid, lead tetra-acetate, peroxyacids and N-bromosuccinimide for allylic bromination. In general, degradative oxidation has been the most powerful tool for elucidating the structures of terpenoids.
- UV spectroscopy has been much used in terpenoid chemistry, its main application being the detection of conjugation.

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- IR spectroscopy is also useful in terpenoid chemistry, and is very valuable for detecting the presence of a hydroxyl group or an oxo group.
- NMR spectroscopy is used to detect the structure of terpenoids.
- Mass spectrometry is now being increasingly used as a means of elucidating the structure of terpenoids. It is possible to determine molecular weights, molecular formulae, the nature of various functional groups and the relative positions of double bonds. It is possible to identify a terpenoid by comparison of its mass spectrum with the reference spectrum of an authentic specimen.

- Optical rotation methods have been successfully applied to the elucidation of the structure of terpenoids, and ORD studies have been used to assign absolute configurations.
- X-ray analysis is very useful, where applicable, for elucidating structure and stereochemistry of terpenoids.
- The final proof of structure depends on synthesis.

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Double Bond Equivalents (D.B.E.)

- The double bond equivalents (D.B.E.), i.e., the number of double bonds and/or rings, in the compound to be ascertained. If the general formula of the compound is $C_aH_bN_cO_d$, then D.B.E. = a + 1 – (b – c)/2.
- Benzene (C_6H_6): D.B.E. = 6 + 1 – 6/2 = 4 (3 double bonds; one ring.
- Allylamine (C_3H_7N): D.B.E. = 3 + 1 – (7 – 1)/2 = 1 (one double bond).

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Monoterpenoids

- The monoterpenoids may be subdivided into three groups: acyclic, monocyclic and bicyclic.

Myrcene

Molecular formula: $C_{10}H_{16}$, b.p. 166-168 °C.
D.B.E. = 10 + 1 – 16/2 = 3

$$\underset{C_{10}H_{16}}{\text{Myrcene}} \xrightarrow[Pt]{H_2} \underset{C_{10}H_{22}}{\text{Decane}}$$

Thus myrcene contains three double bonds and is an open-chain compound.

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Myrcene $\xrightarrow{\text{maleic anhydride}}$ An Adduct

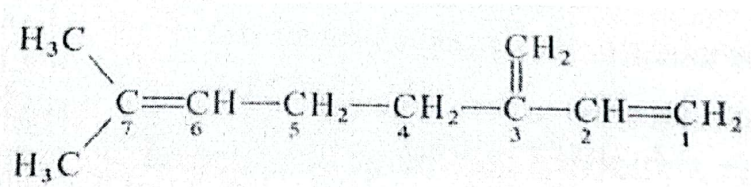
So, two of the double bonds are conjugated. This conjugation is supported by the fact that myrcene shows optical exaltation. It is further proved by UV absorption spectrum (λ_{max} 224 nm).

Myrcene $\xrightarrow{\text{ozonolysis}}$ Acetone + Formaldehyde + A Ketodialdehyde ($C_5H_6O_3$)

Ketodialdehyde $\xrightarrow{\text{oxidation with chromic acid}}$ Succinic acid + CO_2

These results can be explained by assigning structure (I) to myrcene.

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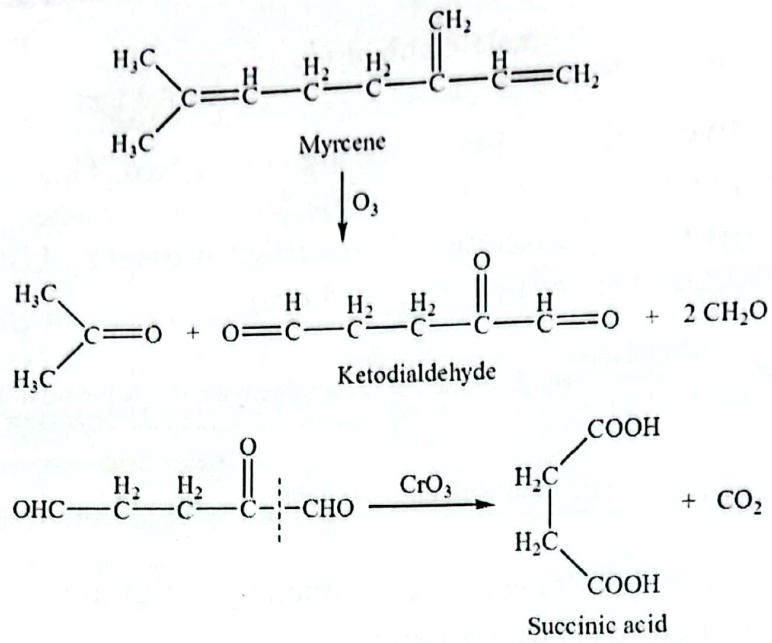


(I)

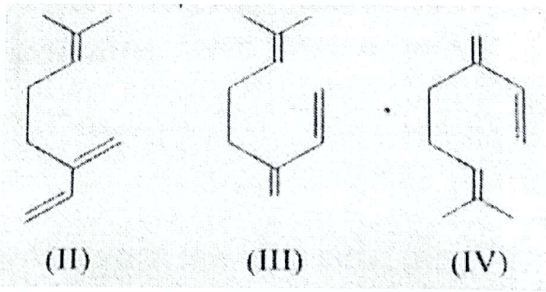
Myrcene
(7-Methyl-3-methyleneocta-1,6-diene)

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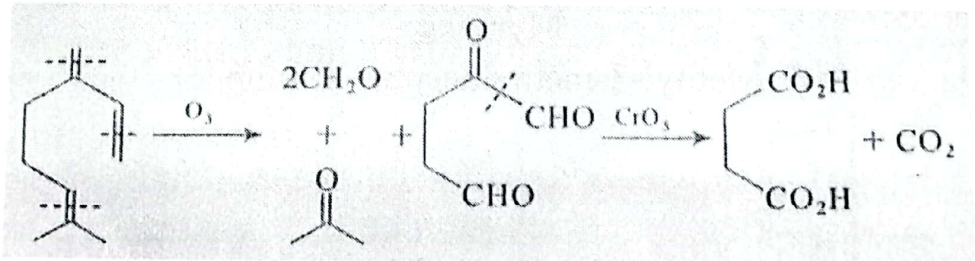
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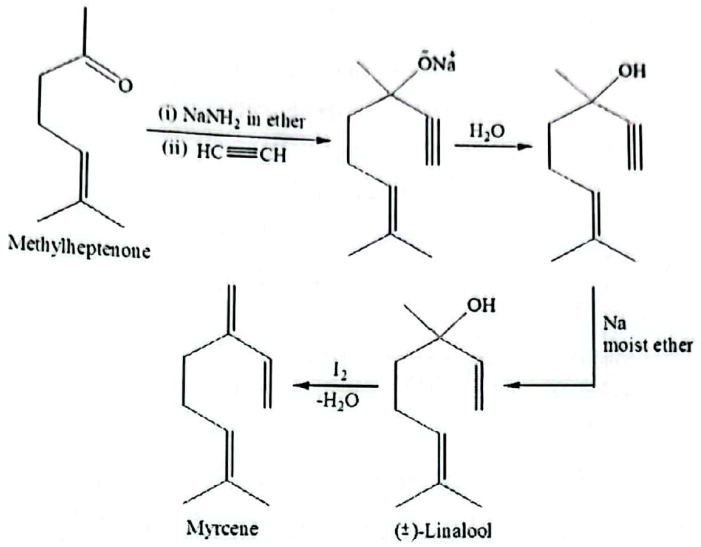


(IV) is recommended.



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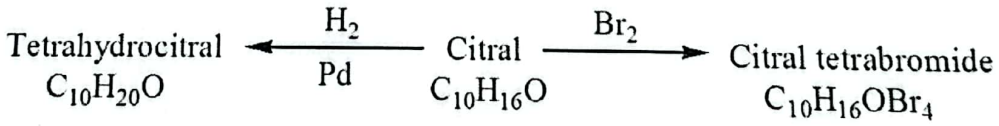
Synthesis



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Citral

- Citral is widely distributed and occurs to an extent of 60-80% in lemon grass oil.
- Citral is a liquid which has the smell of lemons.
- Molecular formula: $C_{10}H_{16}O$
- D.B.E. = 10 + 1 – 16/2 = 3



Citral is found to have two carbon-carbon double bonds.

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Citral (>C=O) $\xrightarrow[\text{(hydroxylamine)}]{H_2NOH}$ an Oxime (>C=NOH)

- It can reduce Fehling's or Tollen's reagent. So, oxygen must be present as an aldehydic group (-CHO).

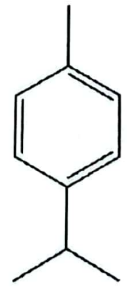
Citral $C_{10}H_{16}O$ (-CHO) $\xrightarrow[\text{reduction}]{Na/Hg}$ Geraniol (a 1^0 alcohol) $C_{10}H_{18}O$ ($-CH_2OH$)

Citral $C_{10}H_{16}O$ (-CHO) $\xrightarrow[\text{oxidation}]{Ag_2O}$ Geranic acid $C_{10}H_{16}O_2$ (-COOH)

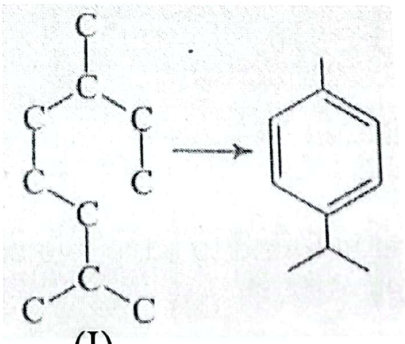
Since there is no loss of carbon on oxidation to the acid, the oxo group in citral is therefore an aldehyde group.

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Citral $\xrightarrow[\Delta]{KHSO_4}$ *p*-Cymene



- This reaction was used by Semmler (1891) to determine the positions of the methyl and isopropyl groups in citral. The citral is acyclic and the skeleton structure is (I).



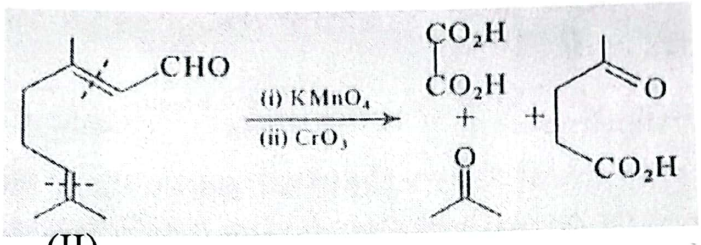
(I)

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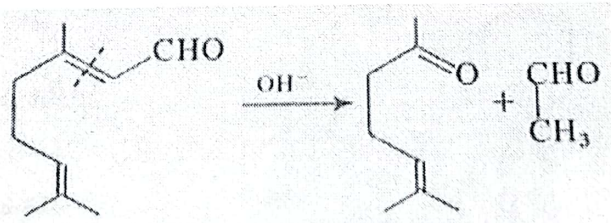
Citral $\xrightarrow[(ii)\ CrO_3]{(i)\ alk.\ KMnO_4}$ Acetone + Oxalic acid + Laevulic acid

- Oxidation of citral with alkaline permanganate, followed by chromic acid gives acetone, oxalic acid and laevulic acid. If citral has structure (II), the formation of these oxidation products may be accounted for.



(II)

- This structure is supported by the fact that aqueous K_2CO_3 converts citral into 6-methylhept-5-en-2-one (III) and acetaldehyde.

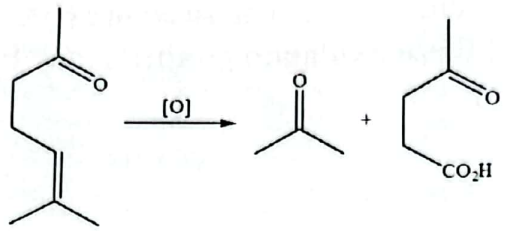


6-Methylhept-5-en-2-one
(III)

This cleavage by alkaline reagents is a general reaction of α,β-unsaturated oxo compounds.

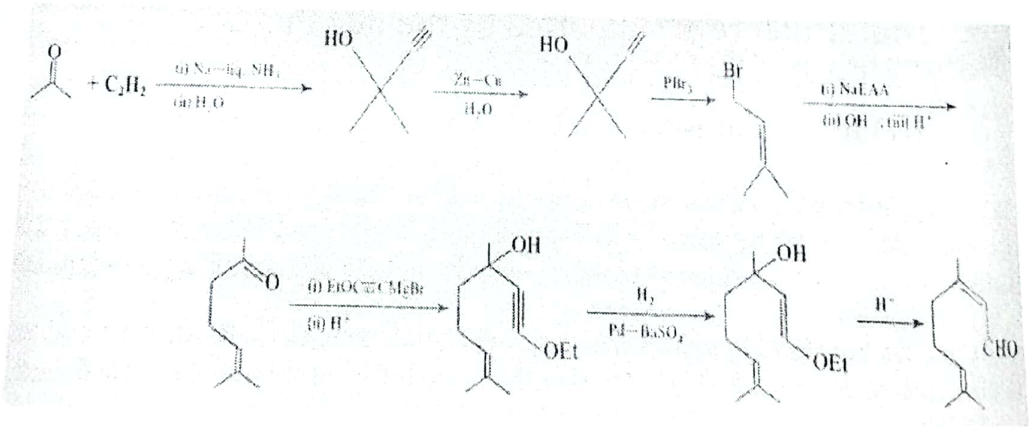
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- Methylheptenone (III) itself is also oxidized to acetone and laevulic acid; this is again in accord with structure (II).



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Synthesis



Citral

$$ClCH_2CH(OC_2H_5)_2 \xrightarrow{NaNH_2} CH{\equiv}COC_2H_5 \xrightarrow{RMgBr} BrMgC{\equiv}COC_2H_5$$

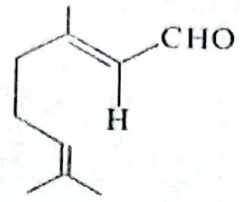
Chloroacetaldehyde diethyl acetal

Ethoxyacetylene-magnesium bromide

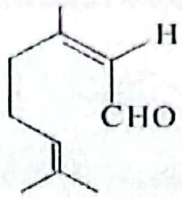
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- Two geometrical isomers are possible for citral. Both isomers occur in natural citral, e.g., two semicarbazones are formed by citral.



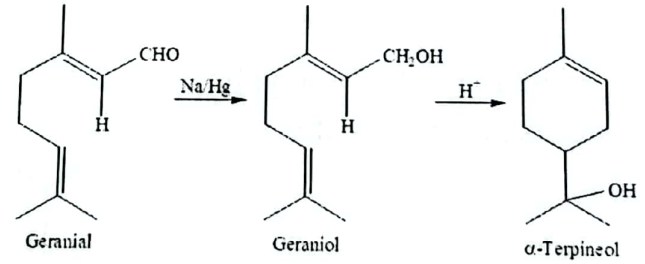
trans- (or *E*-) form; citral-a; geranial



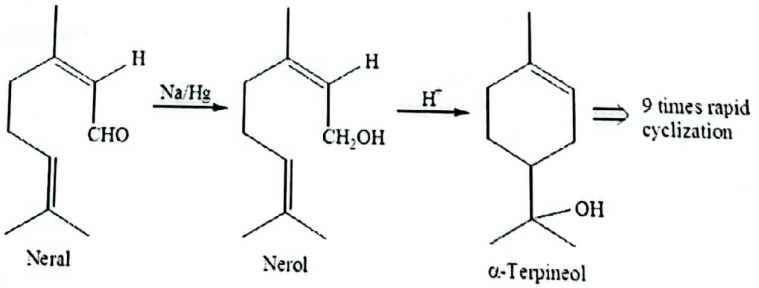
cis- (or *Z*-) form; citral-b; neral

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- The configurations of these two forms have been determined from a consideration of the ring closures of the corresponding alcohols.



Geranial → Geraniol → α-Terpineol

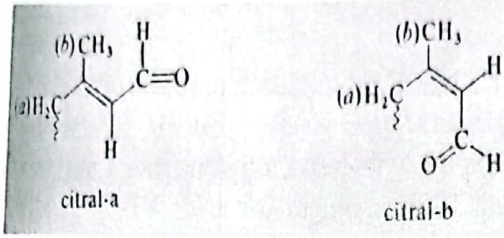


Neral → Nerol → α-Terpineol

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- These assignments have been confirmed by the examination of the NMR spectra of citral-a and citral-b (in $CDCl_3$).

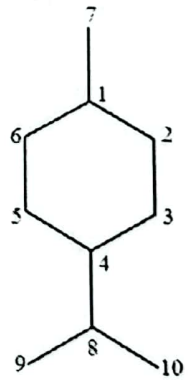


	CH_2 (*a*)	CH_3 (*b*)
citral-a	τ 7·76	τ 7·84
citral-b	τ 7·42	τ 8·02

These happen due to the position of –CHO group (anisotropic effect).

Monocyclic Monoterpenoid

- The parent substance of monocyclic monoterpenoids is *p*-menthane, $C_{10}H_{20}$.



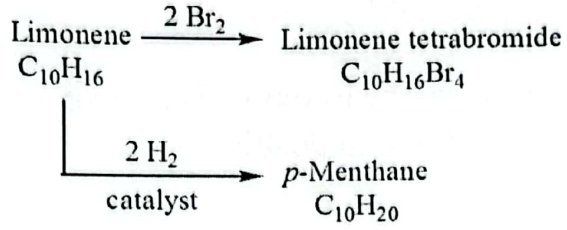
p-Menthane

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Limonene

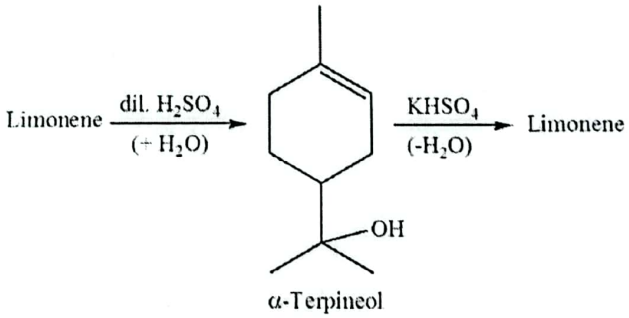
Molecular formula: $C_{10}H_{16}$

- Limonene is optically active; the (+)-form occurs in lemon and orange oils, the (-)-form in peppermint oil, and the (±)-form in turpentine oil. The racemic modification is also known as dipentene.
- Boiling point: 175.5-176.5 °C
- D.B.E. = 10 + 1 – 16/2 = 3

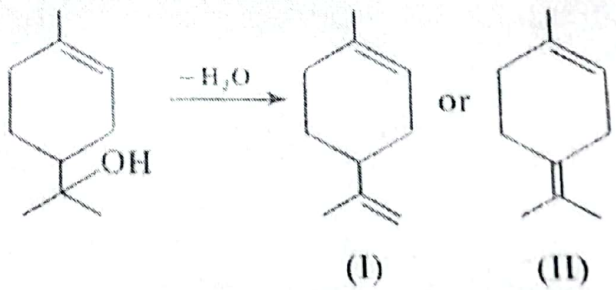


Limonene contains two double bonds and a ring.

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Limonene may be (I) or (II).



Thus the carbon skeleton and the position of one double bond in limonene are known.

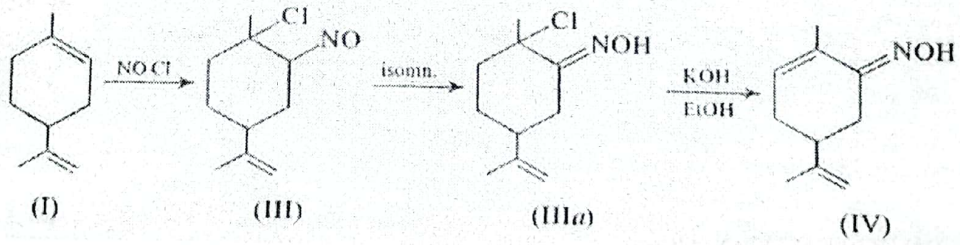
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- Proof of position 8: Sturcture (I) contains a chiral centre C-4, and hence can exhibit optical activity. (II) is symmetric and so cannot be optically active. Therefore (I) must be limonene.
- Chemical proof for position 8 is afforded by the following reactions:

Limonene (I) $\xrightarrow{NOCl}$ Limonene nitrosochloride (III) $\xrightarrow[EtOH]{KOH}$ carvoxime (IV)

Since the structure of carvoxime is known, (I) must have one double bond in position 8.



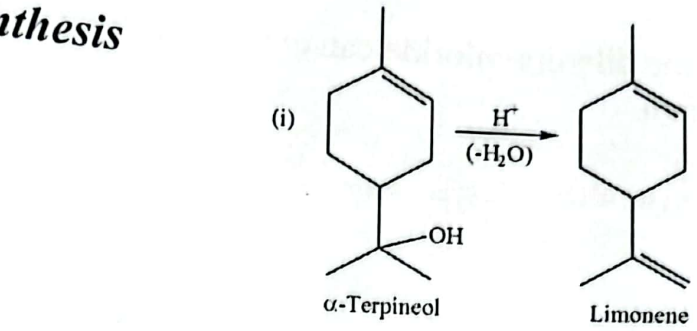
(I) (III) (IIIa) (IV)

Limonene nitrosochloride

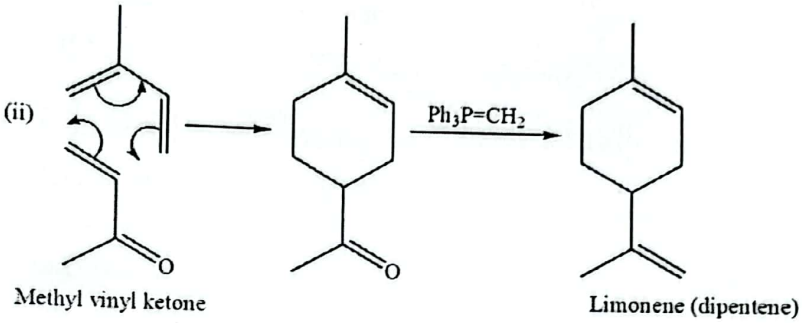
Carvoxime

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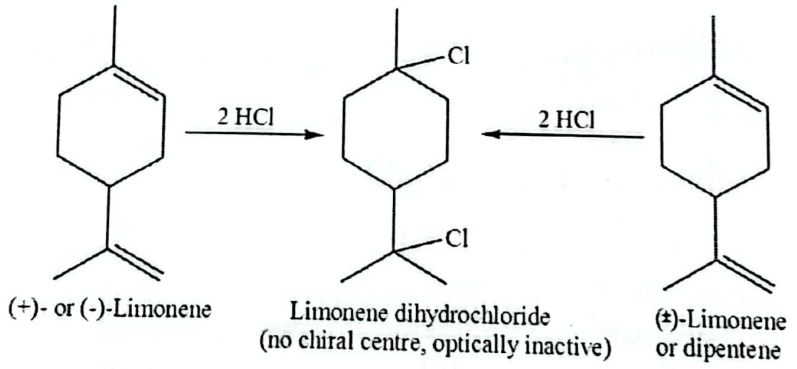
Synthesis



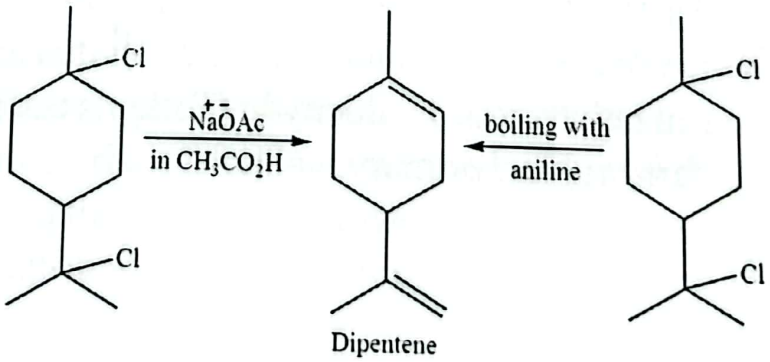
α-Terpineol — Limonene



Methyl vinyl ketone — Limonene (dipentene)



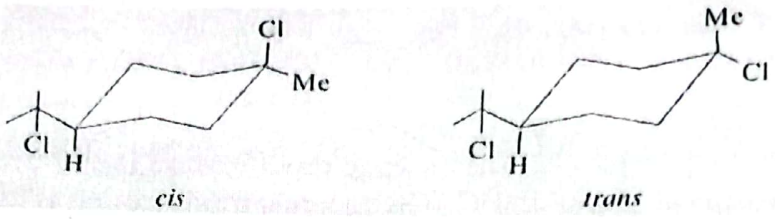
(+)- or (-)-Limonene — Limonene dihydrochloride (no chiral centre, optically inactive) — (±)-Limonene or dipentene



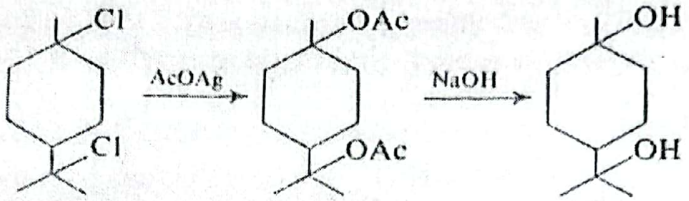
Dipentene

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- Limonene dihydrochloride can exhibit geometrical isomerism.



cis *trans*



1,8-terpin

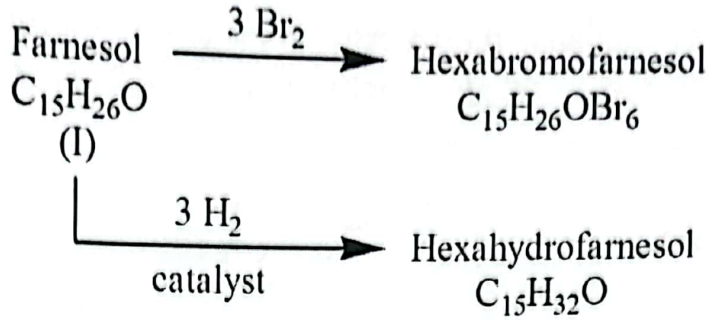
Acyclic Sesquiterpenoids
Farnesol

- Molecular formula: $C_{15}H_{26}O$
- D.B.E. = 15 + 1 – 26/2 = 3

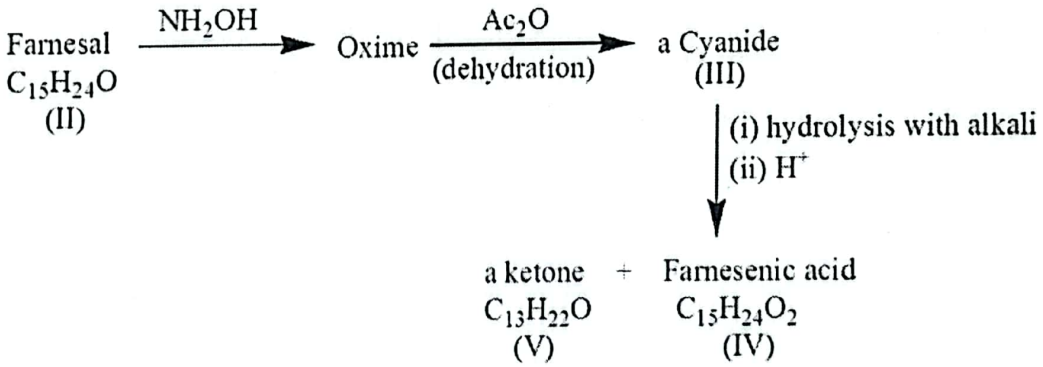
Farnesol $C_{15}H_{26}O$ (I) $\xrightarrow[[O]]{\text{chromic acid}}$ Farnesal $C_{15}H_{24}O$ (II)

Farnesal behaves as an aldehyde (Tollen's test). Thus farnesol is a primary alcohol.

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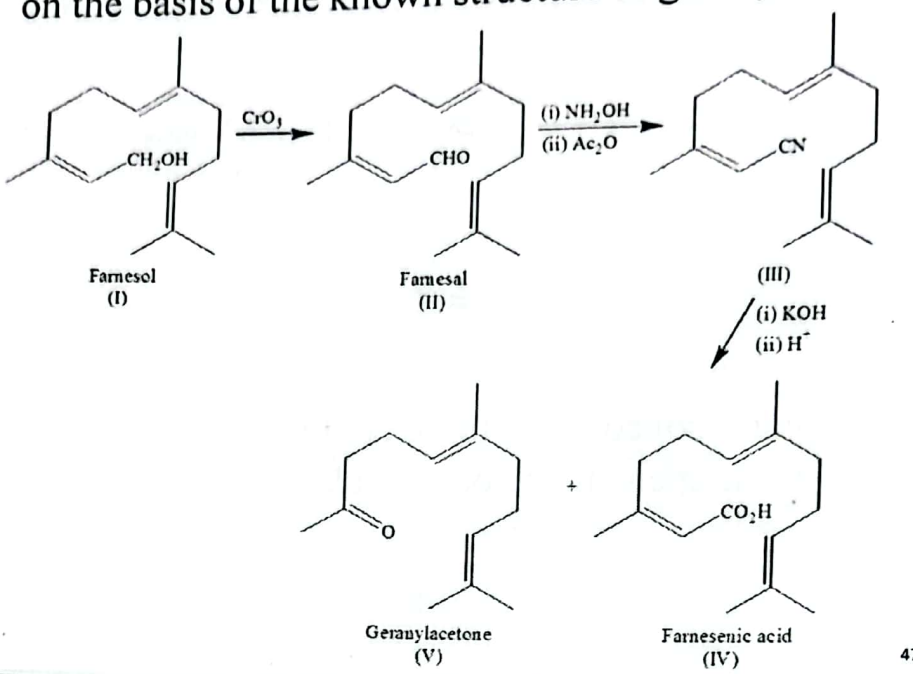
- Farnesol contains three double bonds and it is an open chain compound.



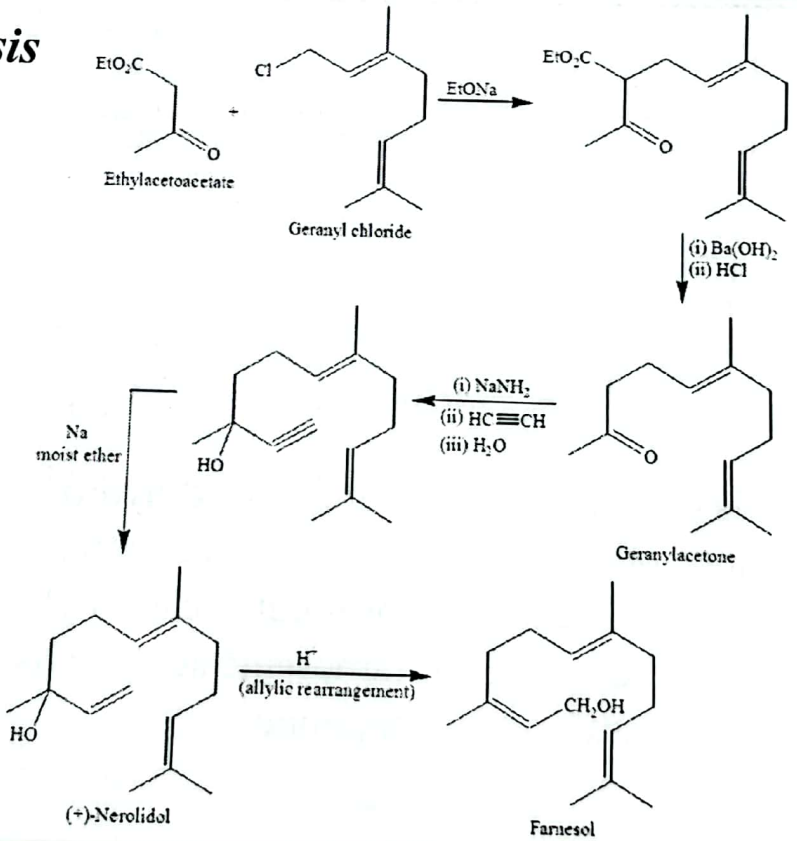
- This ketone (V) was found to be dihydro-*pseudo*-ionone or geranylacetone. This reaction is characteristic of α,β-unsaturated carbonyl compounds and so the precursor, farnesenic acid (or its nitrile) is an α,β-unsaturated compound.

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- Thus the forgoing facts may be formulated as follows on the basis of the known structure of geranylacetone.



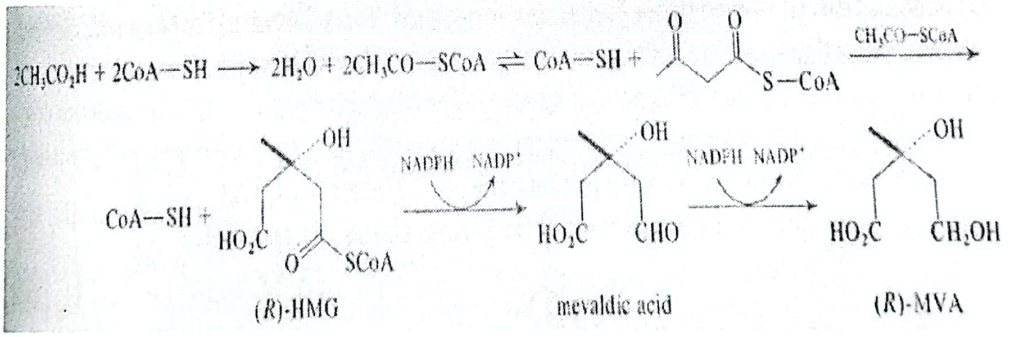
Synthesis



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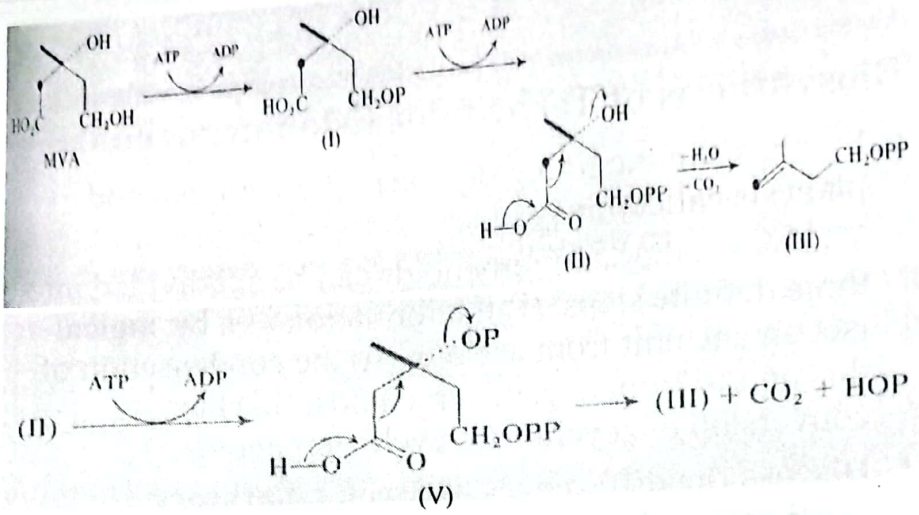
Biosynthesis of Terpenoids (Monoterpenoids)

- Synthesis in the living organism (both animal and plant) is called biosynthesis.
- The biosynthesis of terpenoids can be subdivided into three definite steps: (i) the formation of a biological isopentane unit from acetate; (ii) the condensation of this unit to form acyclic terpenoids; (iii) the conversion of acyclic into cyclic terpenoids.
- **Discuss the differences between laboratory synthesis and biosynthesis.**



HMG: Hydroxymethylglutarate
MVA: Mevalonic acid

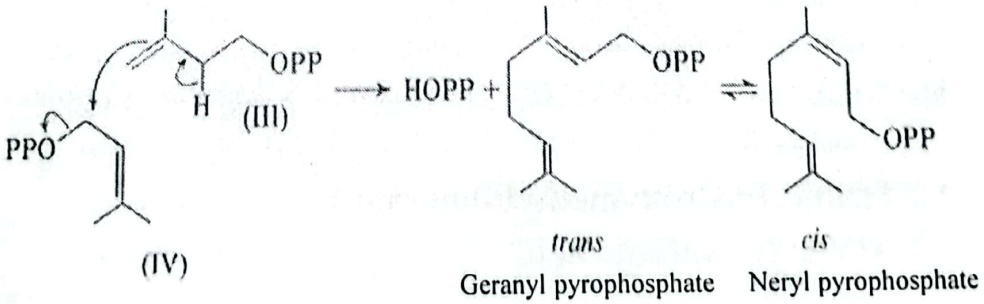
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(I): MVA 5-phosphate; (II): MVA 5-pyrophosphate; (III): 3-Methylbut-3-enyl pyrophosphate or Isopentenyl pyrophosphate; (V): MVA 3-phosphate-5-pyrophosphate; HOP: Phosphoric acid.



(III): 3-Methylbut-3-enyl pyrophosphate
(IV): 3-Methylbut-2-enyl pyrophosphate



Geranyl pyrophosphate Neryl pyrophosphate

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- Geranyl pyrophosphate now serves as the precursor for the monocyclic monoterpenoids via the *cis*-isomer.

